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ORIGINAL ARTICLE

Predicting river system dynamics using machine learning: a multivariable approach to assess the interactions and impacts of hydrological variables on suspended sediment transport

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Abstract
The study of sediment transport in rivers is critical for efficient river management, engineering, and environmental conservation. Neglecting this issue can lead to reservoir sedimentation and ecological degradation. This study focuses on estimating the suspended sediment load (SSL) of the Thoubal River, Manipur, India—a data-scarce and sediment-rich basin—using three machine learning (ML) algorithms: Decision Tree, Random Forest, and Extreme Gradient Boost. A total of 534 datasets, including discharge, suspended sediment concentration, flow area, velocity, flow depth, and flow width, recorded at nine stations between 2021 and 2025, were used as inputs. Beyond prediction, the novelty of this work lies in its comprehensive framework that combines model comparison, uncertainty analysis, and interpretability. First, the models were evaluated using statistical indices (coefficient of determination (R^2), root mean square error (RMSE), overall index of model performance, and ratio of RMSE to standard deviation), where the Random Forest achieved the best performance ($R^2 = 0.98$). Second, Monte Carlo Simulation was applied to verify prediction uncertainty, demonstrating the superior robustness of Random Forest. Finally, two feature analysis methods—feature importance and SHapley Additive exPlanations (SHAP)—were used to identify the relative contribution of hydrological variables, showing that discharge and sediment concentration are the dominant predictors. This integrative approach provides not only accurate predictions but also enhanced interpretability, which is critical for guiding monitoring strategies in data-limited regions. The findings highlight Random Forest as a reliable tool for SSL prediction and suggest its broader application in river basin management, sediment control, and disaster risk reduction.

Keywords Machine learning · Suspended sediment load · Monte Carlo simulation · Random forest · Hydrological variables

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1 Introduction

In hydrological and environmental studies, sedimentation—the process of settling and deposition of eroded soil particles in low lying areas is significant [1]. Water reservoir filling, river course alteration, flood plain expansion, aquatic habitat destruction, and surface water quality deterioration are among the many of the adverse environmental consequences that may arise from sedimentation and sediment accumulation [2–4]. The eroded sediment particles move from upstream to downstream through different transport mechanism. Sediment transport in rivers mainly depends on hydraulic conditions, sediment properties, and channel characteristics. Based on the mode of transport, sediments are generally classified as bedload, which moves along the riverbed by rolling, sliding, or saltation, and suspended load, which is carried within the water column by turbulent flow [5]. A thorough understanding of the processes involved in sediment movement in surface water systems is necessary for forecasting erosion and sediment transport [6]. Researchers are particularly interested in river sediments because of their critical significance in water resources and management strategies; therefore, it is critical to estimate sediment volumes accurately [7–9]. The challenges involved in precisely predicting sediment loads are highlighted by the fact that estimating the amount of suspended sediment in rivers is a complicated procedure impacted by a number of hydrological and climatic parameters [10]. River sedimentation and riverbank erosion in their beds cause sediment deposition in watersheds, which reduces the watershed's flow capacity. Consequently, floodplains may form, raising the likelihood of flooding in the nearby areas [11]. River flow can be greatly impacted and flood damage can increase when rivers contain an excessive amount of sediment [12, 13]. Despite being a critical issue in hydrological studies, sediment movement in hydrodynamic systems is still poorly understood.

The ability to accurately anticipate suspended sediment is crucial for comprehending river morphology and utility water supply issues. Numerous techniques, including direct measurements at sediment monitoring stations, sediment rating curves, regression, artificial intelligence techniques, and empirical approaches based on experimental investigations, can be applied to evaluate the suspended sediment load (SSL). Direct measurements from sediment observation stations are the most accurate approach to determining the sediment material, but due to the sampling procedure, it is expensive, time-consuming, and prone to errors [14, 15]. Sediment- discharge rating curves typically rely on simple empirical power-law relationships. These models assume a stationary, linear and deterministic relationship between discharge and sediment load. However, real river systems are non-linear and non-stationary influenced by rainfall patterns, land use changes and human interventions. Conventional sediment rating curves fail to capture hysteresis, time-lag effects and seasonal variability, leading to inaccurate predictions under dynamic flow conditions [16, 17]. Moreover, conventional methods lack the flexibility to represent dynamic sediment-flow interactions under changing hydrological regimes. They are often based on standard field data and do not generalize flow conditions across varying cross-sections and especially where longitudinal bed profile has been incised due to instream sand mining.

To overcome these limitations, machine learning (ML) techniques have emerged as powerful tools due to their ability to model non-linear and complex relationships [18]. ML algorithms adapt to variations in physical and hydro-meteorological variables using available data and have demonstrated superior performance compared to conventional models

[19, 20]. Numerous studies have successfully applied ML techniques to predict sediment concentration and water quality parameters [21–26]. Hybrid approaches—combining data-driven and physical models—have further enhanced prediction accuracy [27, 28]. Recent works also highlight the effectiveness of artificial neural networks, support vector regression, and hybrid wavelet-based models in SSL estimation [29–33]. However, previous algorithms such as ANN and regression-based approaches still face important shortcomings. ANN models, although powerful, often behave as "black-boxes" with limited interpretability, are highly sensitive to parameter tuning, and risk overfitting in small datasets [34, 35]. Regression-based methods assume linear and stationary relationships, making them unsuitable for capturing the nonlinear and dynamic behavior of sediment transport [36]. In contrast, tree-based ensemble methods offer interpretability, robustness, and flexibility—Decision Trees (DTree) provide simple interpretability but can overfit; Random Forest (RF) reduces overfitting by combining multiple trees [37] and Extreme Gradient Boosting (XGB) further enhances accuracy by employing boosting and regularization [38]. These advantages make DTree, RF, and XGB particularly suitable for SSL prediction in data-scarce and complex hydrological systems.

Building on these advantages, this study applies state-of-art ML algorithms for SSL prediction in the Thoubal River, Manipur, India, a river known for high sediment loads and significant mining activities. Hydro-meteorological parameters—including river discharge (Q), suspended sediment concentration (C_s), velocity (V), flow area (A), flow width (W), and flow depth (D)—were collected from nine sites along the river between 2021 and 2025, and subsequently used to develop three ML models: DTree, RF, and XGB, for SSL prediction. Model performance was evaluated using statistical parameters, and the most influential predictor variables were identified to improve interpretability and guide future monitoring strategies. By combining data-driven approaches with river-specific hydro-meteorological data, this study contributes to a more reliable methodology for SSL prediction in regions where direct measurement is difficult or costly. Furthermore, two feature analysis methods—feature importance and SHapley Additive exPlanations (SHAP)—were applied to assess the contribution of each variable in the model for estimating SSL.

2 Materials and methods

2.1 Study area

The current study focuses on the Thoubal River of Manipur India. Thoubal River is the second largest tributary of the Imphal or Manipur River after Iril River, and is a prominent river system of the Manipur River Basin [39]. The river originates from the steep slope hills of Shirui range (25°07′44″ N and 94°25′09″ E) of Ukhrul District at an elevation of about 1482.00 MSL. The river flows through the hilly terrains from north to south before it enters into plain valley region of Imphal East and Thoubal districts (24°39' N and 93°59' E), where it carries along surmountable sediment and sand load. The river bends towards south-west direction and passes through Yairipok and Thoubal towns before joining the Imphal River at Irong Ichil near Mayang Imphal. Geologically the upstream river reach is composed of the Ophiolite Belt with argillaceous sediment flux derived from a granitic continental crust [40]. The total length of the river is approximately 140.00 km with about 60% of this length flow-

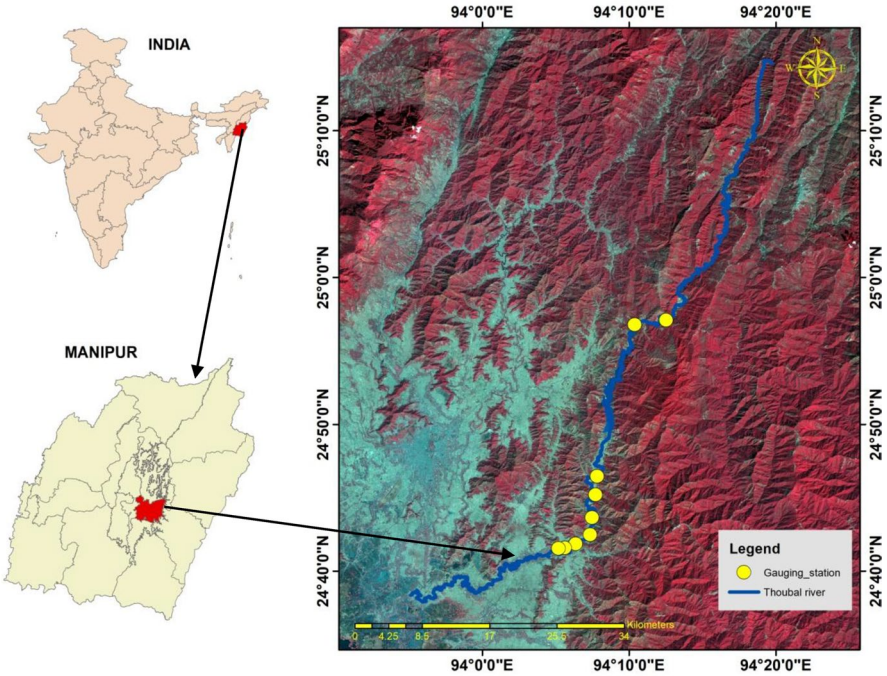

Fig. 1 Location of Thoubal River and gauging stations, Manipur, India

Table 1 Statistical metrics of hydrological data of Thoubal River

Parameter	Statistical metrics				
	X_{min}	X_{avg}	X_{max}	Var	SD
Q (m^3/s)	1.18	25.68	248.98	888.39	29.80
SSL(tons/day)	7.62	581.93	7252.45	666,498.82	816.39
C_s (gm/l)	0.041	0.248	2.31	0.026	0.163
V (m/s)	0.46	1.035	3.2	0.17	0.42
A (m^2)	1.5	23.07	95.76	257.35	16.04
D (m)	0.15	0.69	2.75	0.14	0.38
W (m)	6.4	41.23	94.2	386.73	19.66

Q=river flow, SSL=suspended sediment load, C_s=suspended sediment concentration, V=velocity, A=flow area, D=flow depth, W=flow width, X_{min}=minimum value, X_{avg}=mean value, X_{max}=maximum value, Var=variance, and SD=standard deviation

ing amidst the hilly region and the remaining 40% in the valley region. The study covers a river reach of 47.55 km at upstream station 94.208561° and 24.951381° and downstream 94.086550° and 24.692342°. The location of the gauging stations along the Thoubal river reach undertaken in the study is shown in Fig. 1. Table 1 represents the statistical investigations of the hydrological dataset.

2.2 Data used and preprocessing

In the current study total 534 daily datasets of hydrological parameters (D, V, W, A and C_s) recorded by the authors at an interval of 10 days during the period Nov 2021-Jan 2025 at nine gauging stations as part and partial of the NHP research project implemented at the Department of Civil Engineering, National Institute of Technology Manipur [41, 42], were used to determine the Q and SSL of the river. 70% values of total dataset was used for training and remaining 30% was used for the testing of parameters, consistent with previous hydrological modeling studies [43–45]. Table 1 represents the statistical investigations of the hydrological dataset**.**

Before model development, data normalization was performed to standardise the dependent and independent parameters in the range of 0 and 1. Data normalization ensures that all variables are received equal attention and eliminates their dimension during calibration [46]. Data normalization was performed as follows:

$$\left(x\prime = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \right) \tag{1}$$

where, $x\prime$,x, $x_{\min}$ and $x_{\max}$ represent the normalized, actual, minimum and maximum values of the dataset, respectively.

2.3 Analysis framework

Three different ML algorithms DTree, RF, and XGB using the 'DecisionTreeRegressor', 'RandomForestRegressor', and 'XGBRegressor' functions, respectively, available in the Python framework, were employed in this study to estimate the SSL of the Thoubal river reach under consideration. The findings were then confirmed using several statistical parameters. ML is a significant branch of artificial intelligence that emphasizes on making inferences from existing data and eventually develops algorithms to forecast accurate response variables [47]. Three regression-based ML algorithms- DTree, RF and XGB are employed in this study out of the various approaches that are available in the literature for ML. Although linear regression techniques are also available for ML, they do not consider nonlinear relationships among the variables. Thus, the current study explores the aforementioned ML techniques. For the selection of hyperparameters, sequential parameter optimization was followed, in which different values for each parameter were systematically tested. Each hyperparameter was varied individually while keeping the others fixed at reasonable defaults, and the value yielding the best performance was selected.

Q and C_s, V, D, and W data of each station from 2021 to 2025 were taken into consideration as inputs, and the amount of SSL were computed by each model. The selection of input variables was guided by both hydrological relevance and data availability. Q and C_s were chosen as primary predictors, since they are well-established drivers of SSL dynamics in river systems. These variables were selected based on prior literature [48, 49], which consistently identifies discharge as the dominant control on sediment transport, with sediment concentration reflecting catchment erosional responses. V, D and W were included as secondary predictors to account for hydraulic geometry and channel characteristics influencing sediment-carrying capacity and deposition processes [50, 51].This combination ensured

that the inputs were both scientifically justified and practically reliable for robust model development.

2.4 Modeling approaches

2.4.1 Decision tree (DTree)

DTree, as described by Quinlan [52], is a method for inductive inference that build classification models by recursively partitioning a dataset based on attribute tests. A Decision Tree for regression is a non-parametric supervised learning model that predicts a continuous outcome by recursively splitting the space into smaller and more homogeneous regions, and then fitting a simple regression model within each region. The dataset is split into nodes (internal decision points) and leaves (terminal node). The tree partitions the predictor space into M regions R_1, R_2,……,R_m model the response as a constant C_m in each region:

$$f(x) = \sum_{m=1}^{M} c_m I(x \in R_m) \tag{2}$$

where.

M= number of terminal nodes

c_m =mean response in region R_m

$I(.)$= indicator function.

c_m can be represented as the average of y_i in region R_m based on the criterion minimization of the sum of squares $\sum (y_i - f(x_i))^2$:

$$c_m = ave(y_i | x_i \in R_m) \tag{3}$$

Computationally it is infeasible to find the best binary partition in terms of minimum sum of squares. Hence the algorithm looks for best predictor variable by splitting variable j and split point s, and define the pair of half-planes.

$$R_1(j,s) = \{X | X_j \leq s\} \text{ and } R_2(j,s) = \{X | X_j > s\} \tag{4}$$

The optimal split is chosen by minimizing:

$$\min_{j,s} \left[\min_{c_1} \sum_{x_i \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j,s)} (y_i - c_2)^2 \right] \tag{5}$$

where,

$$c_1 = ave(y_i | x_i \in R_1(j,s)) \text{ and } c_2 = ave(y_i | x_i \in R_2(j,s)) \tag{6}$$

A regression tree must be grown to an appropriate size: if it is too large, it will overfit, while if it is too small, it may miss important structure. Tree size therefore controls model

complexity and should be determined adaptively from the data. Instead of stopping early at weak splits (which might hide useful splits further down), the common strategy is to first grow a large tree until nodes reach a minimum size, and then apply cost-complexity pruning to remove unnecessary branches, producing an optimally sized subtree.

A subtree $T \subset T_0$ is defined to any tree that can be obtained by pruning T_0, that is, collapsing any number of its internal nodes. Terminal nodes are indexed by m, with m representing region R_m. Let $|T|$ denote the number of terminal nodes in T. Letting

$$N_m = |\{x_i \in R_m\}|,$$

$$c_m = \frac{1}{N_m} \sum_{x_i \in R_m} y_i$$

$$Q_m(T) = \frac{1}{N_m} \sum_{x_i \in R_m} (y_i - c_m)^2 \tag{7}$$

Cost complexity criterion is defined as

$$C_\alpha(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T| \tag{8}$$

The parameter α controls the tradeoff between tree size and fit: larger α yields smaller trees, while $\alpha = 0$ gives the full tree. For each $\alpha = 0$ there exists a unique smallest subtree T_α that minimizes the cost function, found using weakest link pruning, which removes nodes that least increase error. This process generates a sequence of subtrees that includes T_α.The optimal α is then chosen via cross-validation, and the final tree is the subtree T_α.

2.4.2 Random forest (RF)

Random forests, as described by Leo Breiman [37], are powerful ensemble learning methods that build a collection of decision trees using random subsets of input features and data, and aggregate their predictions to improve accuracy and stability. Each tree is grown with a unique random vector, ensuring diversity among the trees, and the final prediction is made by majority voting (for classification) or averaging (for regression). Breiman [37] shows that RF do not overfit as more trees are added and that their generalization error depends on the strength of the individual trees and their correlation. Compared to methods like Adaboost, random forests are often equally accurate or superior, faster to train, more robust to noise, and simpler to parallelize. The method also provides internal estimates of error and variable importance using out-of-bag samples, making it a highly practical and interpretable machine learning tool.

In his 2001 work, Breiman [37] introduced several key equations that form the theoretical foundation of the RF algorithm. A central concept is the margin function, which quantifies the confidence of the forest in correctly classifying an input. It is defined as:

$$mg(X,Y) = E_k[I(h_k(X) = Y)] - \max_{j \neq Y} E_k[I(h_k(X) = j)] \tag{9}$$

where, h_k denotes the prediction of the kth tree. Based on this, the generalization error $PE*$of the RF is given by the probability that the margin is negative:

$$PE* = P_{X,Y}(mg(X,Y) < 0) \tag{10}$$

Furthermore, Breiman [37] derived an upper bound for the generalization error, demonstrating that it depends on the strength s of the individual classifiers and the average correlation $\overline{\rho}$ among them, formalized as:

$$PE* \leq \frac{\overline{\rho}(1 - s^2)}{s^2} \tag{11}$$

This relationship emphasizes that minimizing generalization error requires classifiers that are both strong and weakly correlated. In the context of regression, Breiman established a similar convergence result, showing that the mean squared error of the random forest is bounded by the weighted correlation between the residuals of the individual trees.

2.4.3 Extreme gradient boost (XGB)

Chen and Guestrin [38] presented the XGB approach, a novel use of the gradient boosting machine. The "boosting" that forms the basis of XGB can be described as merging the results of multiple "weak learners" to create a "strong learner".

The modified approach has proven to be highly efficient in distributed computing, avoiding overfitting, and resolving issues caused by data sparsity. More specifically, automated learning in the splitting process greatly lowers computing complexity. XGB adds regular terms to the goal function during the learning phase in order to address the overfitting issue. In contrast to traditional gradient boosting, XGB uses second-order Taylor expansion to the loss function to replace the first derivative as shown in equation

$$\mathcal{L}(\varnothing) = \sum_{I} l(\hat{y}_i, y_i) + \sum \Omega(f_k) \tag{12}$$

where l is the mean squared error loss function that measures the difference between the prediction $\widehat{y_i}$ and the observed y_i expressed as $l(y_i, \widehat{y_i}) = (y_i - \widehat{y_i})^2$. Each f_k corresponds to an independent tree structure. The second term Ω penalizes the complexity of the model. where $\Omega(f)$ is calculated as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda w^2 \tag{13}$$

where, T denotes the number of decision trees' leaves, while w denotes scores for each leaf. The γ is a control threshold to split nodes, and λ is a coefficient to reduce over fitting problem. The final equation can be formed as:

$$\widetilde{\mathcal{L}}^{(t)} = \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \tag{14}$$

2.5 Performance metrics

In terms of performance metrics (Equations), the coefficient of determination (R^2), root mean square error (RMSE), overall index of model performance (OI), and ratio of RMSE to standard deviation (RSD) were used, as follows:

$$R^2 = \left\{ \frac{\sum_{i=1}^{N} \left(SSL_{obs} - \overline{SSL_{obs}} \right) . \left(SSL_{pre} - \overline{SSL_{pre}} \right)}{\sqrt{\sum_{i=1}^{N} \left(SSL_{obs} - \overline{SSL_{obs}} \right)^2 \sum_{i=1}^{N} \left(SSL_{pre} - \overline{SSL_{pre}} \right)^2}} \right\}^2 \tag{15}$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{n} \left(SSL_{pre} - SSL_{obs} \right)^2} \tag{16}$$

$$OI = \frac{1}{2} \left\{ 2 - \frac{RMSE}{SSL_{obs}^{\max} - SSL_{obs}^{\min}} - \frac{\sum_{i=1}^{N} \left(SSL_{obs} - SSL_{pre} \right)^2}{\sum_{i=1}^{N} \left(SSL_{obs} - \overline{SSL_{obs}} \right)^2} \right\} \tag{17}$$

$$\mathrm{RSD} = \frac{\mathrm{RMSE}}{\mathrm{STD_{obs}}} = \left[\frac{\sqrt{\sum_{i=1}^{N} \left(SSL_{obs} - SSL_{pre} \right)^2}}{\sqrt{\frac{\sum_{i=1}^{N} \left(SSL_{obs} - \overline{SSL_{obs}} \right)^2}{N}}} \right] \tag{18}$$

where, SSL_{obs} and SSL_{pre} are the observed and predicted SSL values respectively. The variables $\overline{SSL_{obs}}$ and $\overline{SSL_{pre}}$ are the mean of the observed and predicted SSL values, respectively. N indicates the total number of data points.

2.6 Uncertainty analysis

Monte Carlo Simulation (MCS) is a widely used statistical method for estimating the degree of uncertainty in model predictions. Unlike deterministic models that produce a single output, MCS generates a distribution of possible outcomes by repeatedly perturbing input variables or resampling datasets. In each iteration, values of the uncertain inputs are randomly sampled from their underlying probability distributions and the model is executed to produce an output. By repeating this process for a large number of iterations—typically ranging from hundreds to several thousands—a range of predicted values is obtained, which reflects the inherent variability in the system.

Uncertainty in the modeling primarily stems from measurement errors in discharge and sediment concentration, which are particularly challenging in the Thoubal River due to high temporal variability and episodic flood-driven sediment pulses. Additional uncertainty arises from data preprocessing steps, including treatment of missing values, outliers, and normalization, as well as from model-related factors such as algorithmic assumptions and

hyperparameter tuning. In this study MCS was utilized to analyze uncertainty and verify the quality of the findings. Uncertainty in model parameters and input data directly affects predictive accuracy by propagating errors through the modeling process. Inaccuracies in discharge or sediment concentration measurements, or biases introduced during preprocessing (e.g., handling of missing values, normalization), can distort the relationships the model learns, leading to under- or overestimation of suspended sediment load. The accuracy of the model input data determines the uncertainty of the independent variables in this study. To obtain the model outputs, MCS uses random input generation that adheres to a probability distribution [53]. The final step determines the 95% confidence intervals (95PPU-95 Percent Prediction Uncertainty) by taking datasets into account and computing the cumulative distribution's 2.5th (X_L) and 97.5th (X_U) percentiles. The ratio of the observed values that lie within the 95% confidence interval is calculated as follows:

$$95PPU = \frac{1}{k} Count\left(N\left|X_L'' N'' X_U\right|\right) \times 100 \tag{19}$$

in which k denotes the quantity of the observed data points. As the value of SSL decreases (N) and the corresponding X_L and X_U will ascend. The factor d is used to calculate the average width of the confidence interval, as follows

$$d = \frac{\overline{d_x}}{\overline{\sigma_x}} \tag{20}$$

$$\overline{d_x} = \frac{1}{k} \sum\nolimits_{i=1}^{k} \left(X_U - X_L\right) \tag{21}$$

where $\overline{\sigma_x}$ is the standard deviation and $\overline{d_x}$ is the average distance between the upper (97.5th) and lower (2.5th) bands. If $d<1$, uncertainty band is narrower than observed variability, $d=1$ uncertainty band is similar to observed variability and $d>1.5$ the uncertainty band may be too wide.

In MCS, the input dataset comprising Q, V, D, W and C_s was perturbed through bootstrapping. No explicit probability distributions were imposed; instead, the empirical distributions of the observed data served as the source of variability. A total of 100 iterations were performed, generating multiple prediction ensembles from which the 95PPU was derived.

2.7 Sensitivity analysis

In order to determine the sensitivity of input random variables on output three different ML techniques (DTree, RF, XGB) were used to calculate the importance factor of the input variables. Further these results were verified using SHAP (SHapley Additive exPlanations) sensitivity analysis.

SHAP is a game-theoretic method designed to interpret machine learning predictions by attributing output to input features. It has increasingly become a powerful tool in sensitivity analysis, particularly for complex black-box models, where traditional methods like Sobol indices or variance-based decompositions struggle [54]. SHAP sensitivity analysis goes

beyond ranking features—it quantifies how sensitive the output is to each feature in context and supports both global and local interpretations [55, 56].

Core Principles of SHAP for Sensitivity are as follows-

Shapley values: Derived from cooperative game theory, they assess each feature's marginal contribution by averaging over all possible feature combinations.

Global sensitivity: Calculated as the mean absolute SHAP value for each feature across all predictions. This mirrors traditional global sensitivity metrics but is model-agnostic and tailored to the learned data distribution.

Local sensitivity: For individual predictions, SHAP values reveal which features drove that specific outcome.

Dependence and interaction: SHAP supports interaction value decomposition and dependence plots, allowing analysis of conditional feature importance.

3 Results and discussion

The main aim of the current study is to develop ML-based predictive models for evaluating SSL in the considered study reach of Thoubal River and the impact of hydrological parameters on the prediction of SSL. As discussed previously, 30% of the datasets was used for model testing, and 70% of the datasets was used for model training.

Using the default parameter settings of the original ML functions, multiple test runs were performed to explore different parameter combinations. The final sensitivity analysis results for the considered ML models were obtained using the ML parameters summarized in Table 2.

First the performance of the three models in the testing phase in predicting SSL was evaluated employing the evaluation criteria such as R^2, RMSE, OI, and RSD. The considered statistical parameters evaluated for the ML models are tabulated in Table 3.

According to the derived results, the performance of RF model in prediction of SSL outperforms the DTree and XGB models. As shown in Table 3, the RF model with R^2 of 0.988, RMSE of 84.70 (tons/day), OI of 0.98 and RSD of 11%, had the best accuracy in SSL estimations compared to DTree model with $R^2 = 0.938$, RMSE = 191.42 (tons/day),

Table 2 Settings of the parameters for the different ML algorithms

Parameter	Setting		
	DT	RF	XGB
Loss			MSE
Learning rate			0.1
n_estimators		100	100
Criterion	MAE	MSE	friedman_MSE
Max_depth	None	None	3
Min_samples_split	2	2	2
Min_samples_leaf	1	1	1
Max_features	None	None	None
Max_leaf_nodes	None	None	None
Min_impurity_decrease	0	0	0
Alpha			0.9
Validation_fraction			0.1
tol			10^{-4}

Table 3 Comparative performance of presented models for SSL estimation

Model	R^2	RMSE	OI	RSD
DTree	0.938	191.42	0.95	0.26
RF	0.988	84.70	0.98	0.11
XGB	0.978	112.14	0.97	0.15

OI = 0.95 and RSD = 26% and XGB with R^2 = 0.978, RMSE = 112.14 (tons/day), OI = 0.97 and RSD = 15%. The findings of the study were also supported by similar results reported in previous similar studies by different authors. Hanoon et al. [57] conducted a study to predict the suspended sediment load of river systems in Malaysia. In their study the RF model performance metrics R^2 and RMSE were found as 0.83 and 0.031 respectively. In a similar study by Al-Mukhtar [58], for modeling suspended sediment in Tigris River Baghdad, R^2 and RMSE for RF model was found as 0.8 and 130.71 respectively. In comparison, the RF model developed in the present study demonstrated relatively higher predictive accuracy. The superior performance of the RF model in this study can be attributed to three main factors: (i) the use of highly relevant hydrological predictors (Q and C_s) confirmed through feature importance and SHAP analyses; (ii) improved data quality and preprocessing, including min–max normalization and outlier removal; and (iii) careful hyperparameter optimization that balanced bias and variance. The Thoubal River, characterized by highly fluctuating sediment loads and intensive sand-gravel mining activity, exhibits stronger sediment–flow interactions compared to the relatively stable river systems examined by Hanoon et al. [57] and Al-Mukhtar [58]. These pronounced hydro-sediment dynamics enhanced the model's capacity to capture nonlinear relationships between discharge and sediment transport better than that adopted by the two authors. Consequently, the RF model achieved superior statistical performance relative to previous studies, demonstrating that the combination of hydrologically relevant feature selection, effective data normalization, and optimized hyperparameter tuning substantially improves the predictive capability of RF models for SSL estimation. Figure 2 displays a scatter plot and coefficient R^2 comparing the predicted and observed SSL values. The SSL predicted by the RF model had the highest accuracy and was closest to the goodness-of-fit line, as illustrated in Fig. 2b.

3.1 Uncertainty analysis of the selected ML model

This study also employed MCS techniques to evaluate the effectiveness of the model in predicting SSL. The approach was utilized to quantify uncertainties by estimating the confidence intervals of the simulated outputs. Model prediction accuracy was assessed using MCS, the 95% prediction uncertainty range (95PPU), and the d-factor uncertainty index. An acceptable model was introduced since a significant portion of the observed data fell within 95PPU. The findings showed a good match between the 95% confidence intervals and results obtained by RF model with d-factor 0.337 and 90.06% values falling within 95% confidence interval. Therefore, it can be concluded that most of the observed SSL values lay within the confidence intervals. The MCS result of RF model is shown in Fig. 3.

3.2 Sensitivity analysis of the input features

To determine the sensitivity of each input variables in predicting SSL, importance factor were determined using three ML techniques as shown in Fig. 4. From Fig. 4 it can be

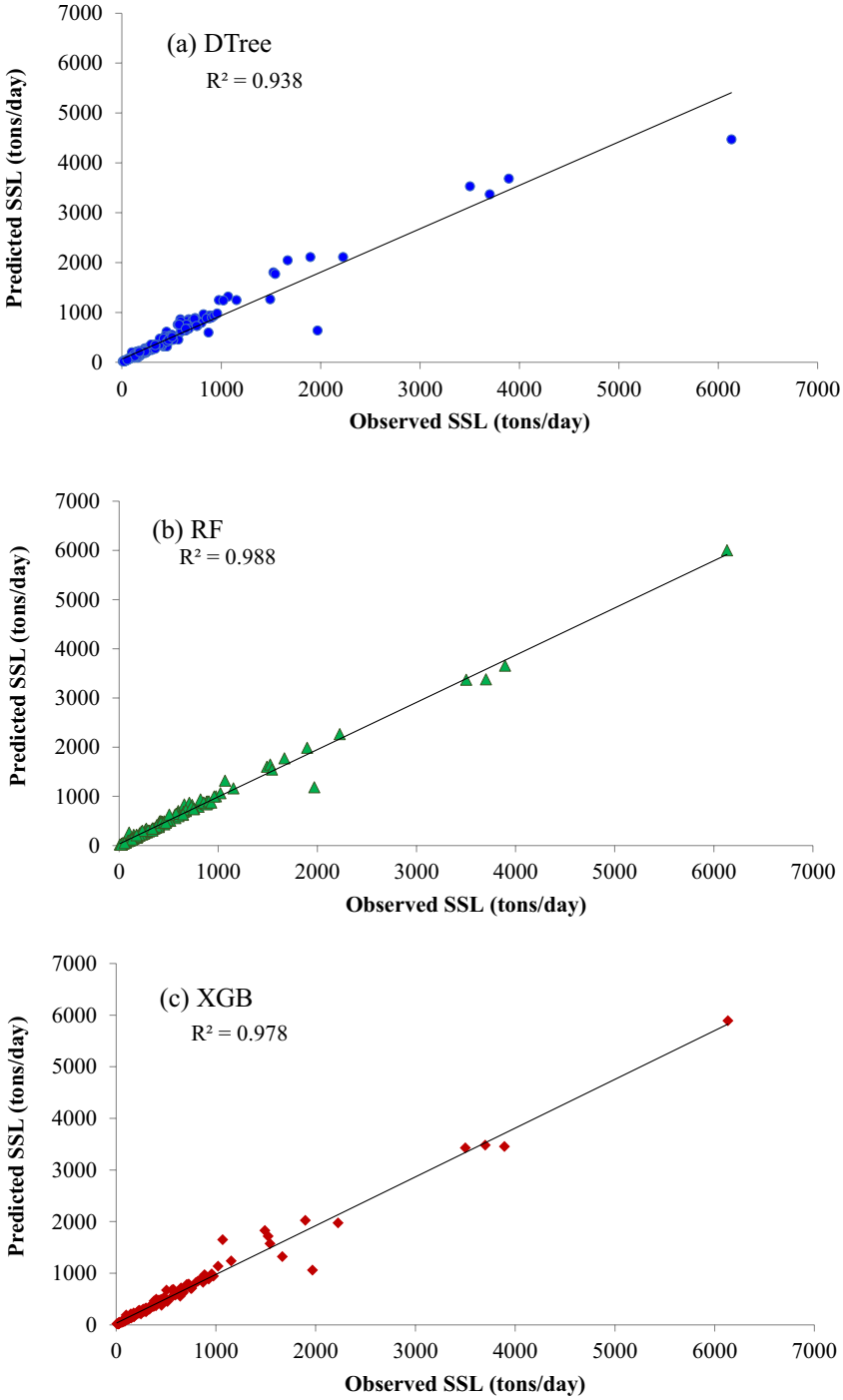

Fig. 2 Observed and predicted suspended sediment load plots for **a** DTree, **b** RF, **c** XGB models

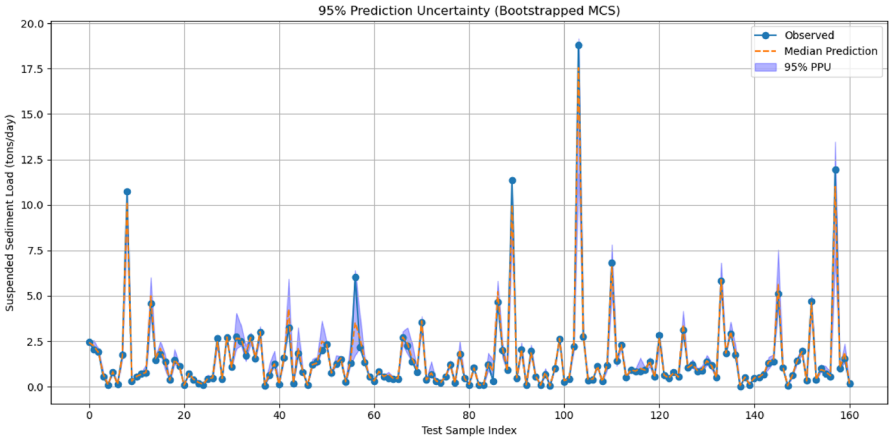

Fig. 3 Uncertainty analysis plot for model performance

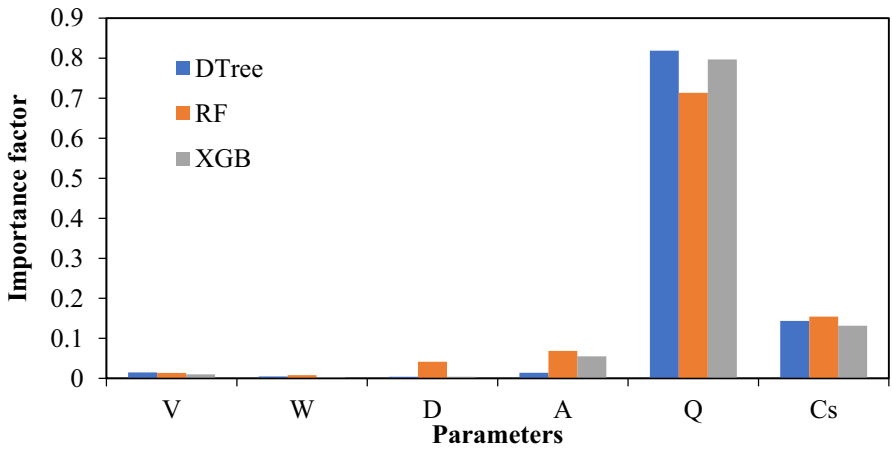

Fig. 4 Importance factor plot for input hydrological parameters

observed that Q has the highest importance factor in predicting SSL as predicted by all three models followed by C_s. The low importance of hydraulic variables (V, A, D, W) in our models can be explained by several interrelated factors. First, Q mathematically integrates velocity and cross-sectional area ($Q = A \times V$), so Q already captures much of the hydraulic information that V, A, D and W would provide; this redundancy reduces the marginal contribution of those variables in feature-importance metrics. Second, these hydraulic variables show limited variance across the sampled events compared with Q and C_s, so they offer little additional signal for predicting SSL. Third, sediment transport is often supply-limited: once entrainment thresholds are exceeded, further changes in local hydraulics produce smaller changes in SSL than changes in available sediment (C_s) or in the magnitude of the flow (Q). Based on the above rationale it can be concluded that a model with Q and C_s will also perform well. As most of the variability in SSL is due to Q and C_s, monitoring programs can

be simplified and resources reallocated: continuous stage/discharge monitoring combined with periodic or automated sampling for C_s will capture the dominant controls on SSL. Hydraulic surveys (cross-sectional area, depth, width) can be maintained at a lower sampling frequency (e.g., seasonal or after major morphological events) to update rating curves and check for regime shifts. Further these results are verified by SHAP sensitivity analysis.

Figure 5 shows the average absolute SHAP values of each feature, reflecting their overall contribution to model predictions. Q (Discharge) has the highest average SHAP value, making it the most influential feature on the model output. This is due to the fact discharge is often a dominant driver in sediment transport, water quality, or hydrological models—it captures the total volume of flow, which typically governs the system's behaviour. C_s (Suspended Sediment Concentration) is the second most impactful variable. Higher C_s often correlates with higher SSL, making it critical for the model to consider. W, A, V and D were found to contribute marginally, suggesting they have a limited influence on the target variable. This can be attributed to redundancy with discharge, since discharge already integrates both velocity and cross sectional area, and to the relatively low variance of these parameters compared to C_s and Q. Either these features are weakly correlated with the target, redundant with stronger predictors, or have limited variation (low entropy), hence the model assigns them minimal weight.

Figure 6 presents the SHAP values for each feature across all data instances, with color indicating the feature's actual value (blue = low, red = high). Q and C_s show a clear spread of SHAP values, indicating they strongly influence both positively and negatively depending on the instance. High Q values (in red) tend to increase the model output, as seen from the cluster of red points with high SHAP values. Low Q values (in blue) are associated with lower or negative impact. This confirms that high Q and C_s values increase the predicted outcome, consistent with their real-world behavior in driving target changes (e.g., sediment load). Other features (A, D, V, W) show compressed SHAP distributions, indicating minor or no variation in influence across samples. This outcome occurs because these parameters show limited variability within the dataset and their effects are incorporated by Q and C_s. Consequently, their predictive power is overshadowed by the dominant influence of Q and C_s.

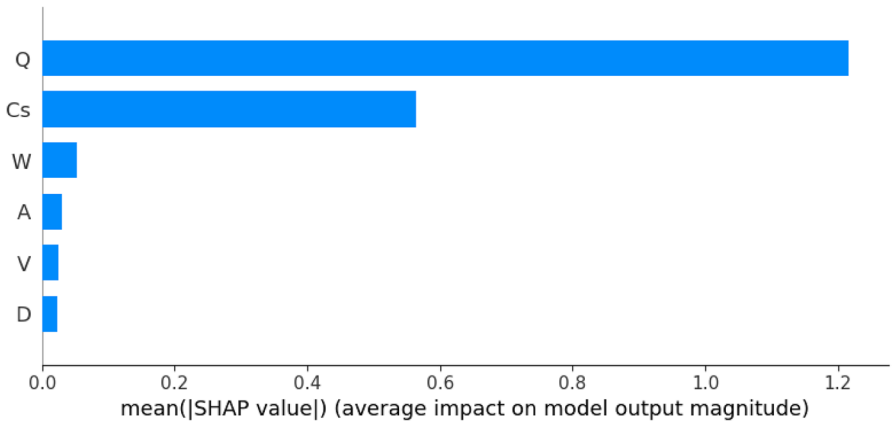

Fig. 5 Mean SHAP values for input hydrological variables

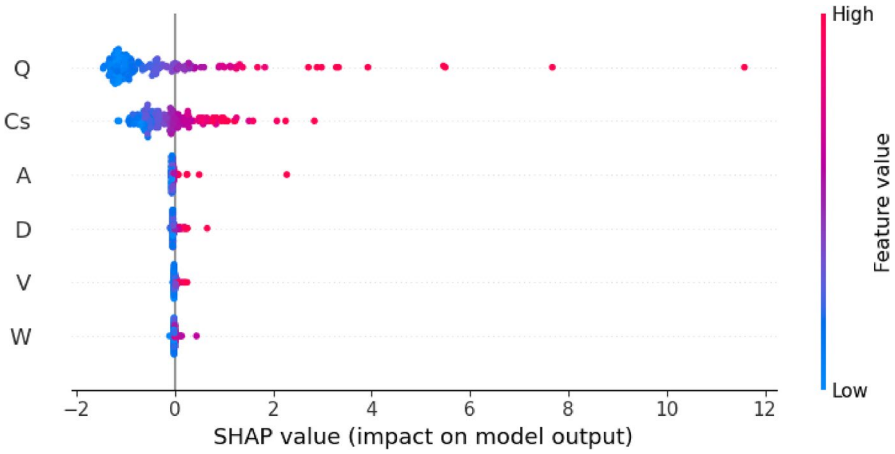

Fig. 6 SHAP values for input hydrological variables

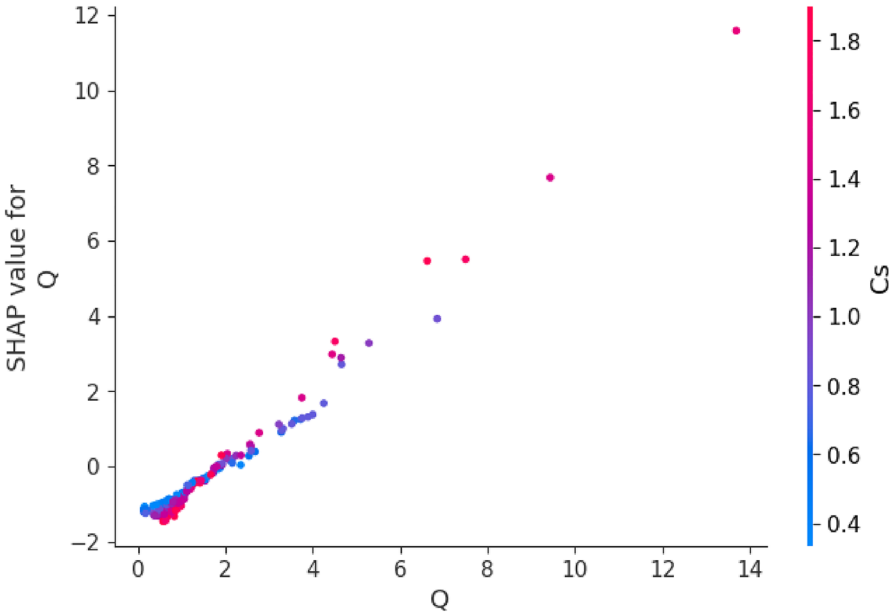

Fig. 7 Variation of actual value of Q with SHAP value of Q and C_s

Relationship between the actual value of Q and its SHAP value, with C_s is shown in Fig. 7. The figure plot indicates how the SHAP value of Q varies with its actual value, colored by the values of C_s. There's a strong positive linear relationship between Q and its SHAP value—as Q increases, its contribution to the model prediction increases. This implies the model treats higher flow as a strong signal for higher output, which aligns with typical hydrological or transport models where Q is a direct driver. Redder points (higher C_s) amplify the SHAP value of Q. This suggests feature interaction—i.e., Q has a greater

positive effect when C_s is also high. It indicates the model learned that a combination of high flow and high sediment concentration is especially predictive of the target outcome (e.g., erosion, deposition, pollutant load). The finding that Q emerged as the most important predictor is expected and consistent with sediment transport theory, as river flow typically governs the capacity to mobilize and carry suspended material. However, the SHAP analysis provides added value by revealing how Q interacts with C_s in shaping the model output. Specifically, the dependence plot (Fig. 7) shows that the contribution of Q to SSL predictions is amplified when C_s is high, highlighting a strong interplay between Q and C_s. This interaction reveals that during periods of high Q, the river has greater energy to mobilize material, but its effectiveness depends on the concurrent availability of sediment. When both Q and C_s are high, typically during monsoon-driven flood events, SSL increases sharply, leading to high turbidity, accelerated channel erosion, and downstream siltation. Conversely, high flows under low C_s conditions resemble "clear water" events, where transport capacity exceeds sediment supply. By uncovering this interaction, SHAP analysis enriches traditional feature ranking approaches and underscores the dual importance of hydrological forcing and sediment availability in regulating water quality and sediment dynamics in the Thoubal River.

4 Conclusion

SSL is a critical water quality parameter that influences water clarity, turbidity, and color. It also plays a vital role in the planning, design, and management of water supply systems and infrastructure. Due to the complexity of the SSL phenomenon, employing machine learning techniques may lead to precise prediction of SSL. In this study, three ML models, namely DTree, RF, XGB were utilised to predict the SSL. The model performance was evaluated based on the statistical parameters such as R^2, RMSE, OI, and RSD. The results show that RF model outperforms the others, yielding lowest RMSE and RSD of 84.7 and 0.11 respectively with highest R^2 of 0.988. The accuracy (OI) of the RF model (0.98) was higher than DTree and XGB models.

To check the uncertainty of the model in SSL prediction MCS was conducted for RF model. The results indicated 90.06% observed value fall within 95% confidence interval with d-factor of 0.337 showing model predicts the SSL within acceptable limits.

To determine the impact of hydrological parameters on SSL prediction importance factor is calculated using the three ML techniques. The results showed that discharge (Q) is the most influential parameter followed by sediment concentration (C_s). Further SHAP sensitivity analysis reveals that the model's predictions are most sensitive to Q and C_s. These features not only show the highest individual contributions but also exhibit interaction effects. In contrast, insignificant influence on the SSL was observed for several features such as W, A, D, and V due to redundancy or weak correlation with the target variable. The model's behaviour aligns well with domain knowledge in hydrological and environmental systems. This study was based on a single 70:30 train–test split; however, future research could adopt cross-validation across multiple basins to further enhance the robustness and generality of the results.

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Author contributions Mukesh Kumar Yadav: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Software, Validation, Resources, Writing – original draft. Ngangbam Romeji: Conceptualization, Supervision, review & editing.

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Data availability The data used in this study were obtained from field measurements along the Thoubal River between 2021 and 2025. The processed dataset and model implementation codes (Decision Tree, Random Forest, and XGBoost) are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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